

MANIPAL UNIVERSITY JAIPUR
Department of Mechanical Engineering
Course Hand-out

A. Basic Details:

Programme Name:	B.Tech.
Course Name:	Engineering Economics
Course Code:	MEE2003
LTPC (<i>Lecture Tutorial Practical Credits</i>):	3 0 0 3
Session:	2026-27
Class:	All Engineering Branches
Course Coordinator:	Dr Ramanpreet Singh
Course Instructor(s):	
Additional Practitioner(s) – if any (<i>Industry Fellow/ Visiting Faculty/ Adjunct Faculty, etc.</i>):	

B. Introduction: The Department of Mechanical Engineering offers the course Engineering Economics, where students from a wide range of disciplines can gain knowledge of economics as it applies to the engineering profession. This course would help students by teaching them the basics of economics and cost analysis as they relate to engineering projects. By learning how to assess cost and income data and make economic decisions, students will be better equipped to defend or reject alternatives and projects based on sound economic reasoning after finishing this course.

C. Course Outcomes:

CO Statement	CO	Bloom's Level	Target Attainment %	Target Attainment level
Describe various economic decision-making concepts and types of estimates.	MEE2003.1	L2	75	2
Assess the financial viability of engineering projects.	MEE2003.2	L3	80	3
Discuss budget and budgetary control concepts.	MEE2003.3	L2	75	2

Analyse for replacement analysis and break-even analysis.	MEE2003.4	L4	80	3
Estimate the projects for economic feasibility by risk analysis to enhance entrepreneurship skills.	MEE2003.5	L3	75	2

D. Program Outcomes and Program Specific Outcomes:

- [PO.1]. Engineering Knowledge:** Apply knowledge of mathematics, natural science, computing, engineering fundamentals and an engineering specialization as specified in WK1 to WK4 respectively to develop the solution of complex engineering problems.
- [PO.2]. Problem analysis:** Identify, formulate, review research literature and analyze complex engineering problems reaching substantiated conclusions with consideration for sustainable development. (WK1 to WK4)
- [PO.3]. Design/development of solutions:** Design creative solutions for complex engineering problems and design/develop systems/components/processes to meet identified needs with consideration for public health and safety, whole life cost, net zero carbon, culture, society and environment as required. (WK5)
- [PO.4]. Conduct investigations of complex problems:** Conduct investigations of complex engineering problems using research-based knowledge including design of experiments, modelling, analysis and interpretation of data to provide valid conclusions. (WK8)
- [PO.5]. Engineering Tool Usage:** Create, select and apply appropriate techniques, resources and modern engineering & IT tools, including prediction and modelling, recognizing their limitations to solve complex engineering problems. (WK2 and WK6)
- [PO.6]. The Engineer and The World:** Analyze and evaluate societal and environmental aspects while solving complex engineering problems for its impact on sustainability with reference to economy, health, safety, legal framework, culture and environment. (WK1, WK5, and WK7)
- [PO.7]. Ethics:** Apply ethical principles and commit to professional ethics, human diversity and inclusion; adhere to national & international laws. (WK9)
- [PO.8]. Individual and collaborative team work:** Function effectively as an individual, and as a member or leader in diverse/ multi-disciplinary teams.
- [PO.9]. Communication:** Communicate effectively and inclusively with the engineering community and society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations considering cultural, language, and learning differences.
- [PO.10]. Project management and finance:** Apply knowledge and understanding of engineering management principles and economic decision-making and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.
- [PO.11]. Life-long learning:** Recognize the need for, and have the preparation and ability for i) independent and life-long learning ii) adaptability to new and emerging technological and iii) critical thinking in the broadest context of technological change. (WK8)

Program Specific Outcomes

[PSO.1].

Model and analyse of Mechanical Engineering systems and components.

[PSO.2].

Design and performance evaluation of energy systems.

[PSO.3].

Testing of materials and analysis of manufacturing processes for mechanical engineering applications.

E. Assessment Plan:

Criteria	Description	Maximum Marks
Internal Assessment (Summative)	Mid-Term Examination	30
	Class Work Sessional (CWS): Quiz, Assignment	30
End Term Exam (Summative)	End Term Exam	40
	Total	100

F. Syllabus: Concept and Value Analysis, Economic Decision Making, Types of Estimates, Accounting and Control, Elements of Cost, Prime Cost, Overheads, Types of Cost, Process Cost & Cost of Production, Break Even Analysis, Inventory Control & Management, EOQ, Financial Analysis, Simple payback, Return on Investment, NPV (Net Present Value), IRR (Internal rate of Return), Life Cycle Cost Method, Sensitivity Analysis, Project Financing Options. Budget and Budgetary Control, Concept of Budgeting, Type of Budgets. Risk - Risk vs Return, System Concept and Value Analysis, System Analysis & System Engineering, Value Analysis. Replacement Analysis, Depreciation.

Text Books

1. R. Panneerselvam, Engineering Economics, Prentice Hall of India.
2. J.L. Riggs, D.D. Bedworth and S.U. Randhawa, Engineering Economics, McGraw Hill Education.
3. P.L. Mehta, Managerial Economics, Sultan Chand & Sons.

Reference Books

1. E.L. Grant, W.G. Ireson and R.S. Leavenworth, Principles of Engineering Economic Analysis, John Wiley.
2. G.J. Tiesen, W.J. Fabrycky and H.G. Tiesen, Engineering Economy, Prentice Hall of India.
3. L. Blank and A. Tarquin, Engineering Economy, McGraw Hill Education.

G. Lecture Plan: The lecture plan of the course indicates clear schedule of content delivery, session outcome, corresponding CO, mode of delivery, and mode of assessment. The general template of lecture plan is mentioned as follows:

Lecture No.	Topic	Session Outcome	Corresponding CO	Mode of delivery	Mode of assessing CO
1	Introduction to Engineering Economics- Nature, scope, basic problems of an economy	Understand the fundamental concepts of Engineering Economics	MEE2003.1	Lecture based teaching-learning	Quiz Assignment Mid Term End Term
2	Introduction to Engineering Economics- Microeconomics and macroeconomics	Understand macroeconomic indicators like GDP, inflation, interest rates, and their impact on engineering projects.	MEE2003.1	Lecture based teaching-learning	Quiz Assignment Mid Term End Term
3	Theory of demand and supply	Understand the fundamental concepts of demand and supply in the context of engineering economics	MEE2003.1	Lecture based teaching-learning	Quiz Assignment Mid Term End Term
4	Demand and supply analysis	Understand the fundamental concepts of demand and supply in the context of engineering economics	MEE2003.1	Lecture based teaching-learning	Quiz Assignment Mid Term End Term
5	Demand and supply analysis (Numerical Problems)	Understand the fundamental concepts of demand and supply in the context of engineering economics	MEE2003.1	Lecture based teaching-learning	Quiz Assignment Mid Term End Term
6	Elasticity of Demand	Understand the concept of elasticity of demand and its	MEE2003.1	Lecture based teaching-learning	Quiz Assignment Mid Term End Term

		various types – price, income, and cross elasticity			
7	Elasticity of Demand (Numerical Problems)	Solve numerical problems related to price, income, and cross elasticity of demand	MEE2003.1	Lecture based teaching-learning	Quiz Assignment Mid Term End Term
8	Economic Decision Making	Understand the principles and importance of economic decision making in engineering and managerial contexts	MEE2003.1	Lecture based teaching-learning	Quiz Assignment Mid Term End Term
9	Importance of Economic Decision Making in Engineering Projects	Understand the role of economic decision making in the successful planning and execution of engineering projects	MEE2003.1	Lecture based teaching-learning	Quiz Assignment Mid Term End Term
10	Types of Estimation- ROM, Budget Estimate	Understand the importance of cost estimation in engineering economics and project planning	MEE2003.1	Lecture based teaching-learning	Quiz Assignment Mid Term End Term
11	Definitive Estimate, Parametric Estimate	Understand the concepts and significance of Definitive and Parametric Estimates in engineering project planning	MEE2003.1	Lecture based teaching-learning	Quiz Assignment Mid Term End Term
12	Analogous Estimate, Bottom-Up Estimation	Understand the concepts of Analogous Estimation and Bottom-Up Estimation in project cost planning.	MEE2003.1	Lecture based teaching-learning	Quiz Assignment Mid Term End Term

13	Estimation Models- EOQ	Calculate the EOQ and understand its application in inventory management.	MEE2003.2	Lecture based teaching-learning	Quiz Assignment Mid Term End Term
14	Simple Payback and rate of return	Understand and calculate the simple payback period for investments.	MEE2003.2	Lecture based teaching-learning	Quiz Assignment Mid Term End Term
15	Time value of money	understand its significance in evaluating investments.	MEE2003.2	Lecture based teaching-learning	Quiz Assignment Mid Term End Term
16	Net Present Value (NPV)	Calculate NPV and understand its use in financial decision-making.	MEE2003.2	Lecture based teaching-learning	Quiz Assignment Mid Term End Term
17	NPV (Numerical Problems)	Calculate NPV and understand its use in financial decision-making.	MEE2003.2	Lecture based teaching-learning	Quiz Assignment Mid Term End Term
18	Internal Rate of Return (IRR)	Determine IRR and its role in comparing investment opportunities.	MEE2003.2	Lecture based teaching-learning	Quiz Assignment Mid Term End Term
19	IRR (Numerical Problems)	Determine IRR and its role in comparing investment opportunities.	MEE2003.2	Lecture based teaching-learning	Quiz Assignment Mid Term End Term
20	Cost Analysis- Concepts of Cost, Classification of Cost, Elements of Costs	Recognize the various elements that constitute cost in production	MEE2003.3	Lecture based teaching-learning	Quiz Assignment Mid Term End Term
21	Life Cycle Cost Methods, Project Financing Options	Analyze costs using the life cycle cost method.	MEE2003.3	Individual learning/ self-study	Quiz Assignment Mid Term End Term
22	Break-Even Analysis	Perform break-even analysis and understand its implications	MEE2003.4	Lecture based teaching-learning	Quiz Assignment Mid Term End Term

23	BEA (Numerical Problems)	Perform break-even analysis and understand its implications	MEE2003.4	Lecture based teaching-learning	Quiz Assignment Mid Term End Term
24	BEA (Numerical Problems)	Perform break-even analysis and understand its implications	MEE2003.4	Lecture based teaching-learning	Quiz Assignment Mid Term End Term
MID TERM EXAM					
25	Revisit entire syllabus till mid-term.		MEE2003.3	Lecture based teaching-learning	Quiz Assignment End Term
26	Budget and Budgetary Control	Understand the concepts of budgeting and budgetary control.	MEE2003.3	Individual learning/self-study	Quiz Assignment End Term
27	Concept of Budgeting	Learn different types of budgets and their applications.	MEE2003.3	Individual learning/self-study	Quiz Assignment End Term
28	Type of Budgets	Analyze various types of budgets and their purposes.	MEE2003.5	Individual learning/self-study	Quiz Assignment End Term
29	Risk and Risk Vs Returns	Understand the relationship between risk and return in financial decisions.	MEE2003.5	Individual learning/self-study	Quiz Assignment End Term
30	Depreciation Methods- Straight line method (Numerical Problems)	Calculate depreciation and understand its impact on financial statements.	MEE2003.5	Lecture based teaching-learning	Quiz Assignment End Term
31	Decline balancing method and double decline balancing method of depreciation (Numerical Problems)	Calculate depreciation and understand its impact on financial statements.	MEE2003.5	Lecture based teaching-learning	Quiz Assignment End Term
32	Sinking fund method of depreciation (Numerical Problems)	Calculate depreciation and understand its impact on	MEE2003.5	Lecture based teaching-learning	Quiz Assignment End Term

		financial statements.			
33	Review of all Depreciation Methods	Review key concepts related to depreciation		Lecture based teaching-learning	Quiz Assignment End Term
34	Concept of Value Analysis	Understand the concept of value analysis	MEE2003.5	Lecture based teaching-learning	Quiz Assignment End Term
35	Phases of Value Analysis	Analyse the phases such as blast, create, and refine	MEE2003.5	Lecture based teaching-learning	Quiz Assignment End Term
36	Review and Assessment	Review key concepts and assess understanding through discussions and quizzes.		Lecture based teaching-learning	
End Term Examination					

H. Course Articulation Matrix:

[illegible]

[illegible]